Detailed Solution Write-up

Title: Physics-Informed Tabular Machine Learning for Spatial-Temporal Urban Flood Routing

Competition: UrbanFloodBench: Flood Modelling

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Date: March 17, 2026

1. Abstract & High-Level Philosophy

Overview of the Approach

This solution formulates the spatial-temporal problem of urban flood modeling through a paradigm of Physics-Informed Tabular Machine Learning. While the prevailing trend in modern sequence and graph-based tasks utilizes deep learning architectures (e.g., Graph Neural Networks or LSTMs), I opted for a robust gradient boosting framework (LightGBM). By embedding fundamental hydraulic principles—such as mass balance, flow accumulation, and pipe capacity approximations—directly into the feature engineering pipeline, the tree-based algorithms were constrained to approximate physical laws rather than merely memorize sequential patterns.

Rationale Against Autoregressive Deep Learning

The decision to bypass autoregressive neural networks (like LSTMs) was driven by three structural vulnerabilities inherent to deep learning in fluid dynamics simulations without strict boundary conditions:

1. **Autoregressive Error Accumulation:** LSTMs predict the state at time t based on the prediction at $t-1$. In a prolonged storm event, a minor error in water level prediction at early stages compounds recursively, potentially leading to physically impossible anomalies during later timesteps. This tabular approach predicts the state independently based on global storm context, eliminating cascading temporal errors.
2. **Risk of Node-Specific Overfitting:** Utilizing node-specific projections in neural networks often leads to memorizing the exact topology and capacity of the training set. Such models exhibit brittleness when exposed to unseen nodes or out-of-distribution weather events.
3. **Interpretability Deficit:** Deep learning models function primarily as black boxes. In contrast, this LightGBM approach provides high feature interpretability, allowing for explicit tracking of how variables like 1D-2D cross-coupling or capacity utilization drive final predictions.

Empirical Validation via Dataset Update

The stability of this physics-informed tabular approach was empirically validated during the competition. When the organizers identified and corrected an error in the dataset mid-competition, many top-ranking models experienced a significant drop in their Public Leaderboard scores, suggesting their architectures had memorized flawed data patterns. Conversely, my LightGBM model was unaffected by the flawed data, and retraining on the corrected dataset yielded a slight score improvement. This indicates that the model successfully generalized the underlying fluid dynamics rather than overfitting to dataset noise.

2. Target Formulation & Normalization

A fundamental challenge in spatial-temporal hydraulic modeling using tabular data is the high variance in absolute target scales. Different nodes (e.g., small 1D pipes versus expansive 2D surface basins) possess entirely different baseline elevations and volumetric capacities. To address this, the learning objective was shifted from absolute value prediction to scale-invariant residual prediction.

Prediction Target (Residual Learning)

Instead of predicting the raw $water_level$, the model was trained to predict the dynamic residual: $target - node_mean$. By subtracting the historical mean water level of each specific node, the target distribution was centered around zero. This transformation decoupled static topographical features from the dynamic hydrological response, requiring the model to only learn water level fluctuations relative to rainfall stimuli.

Initial State Anchoring & Fallback Strategy

To model water accumulation accurately, the algorithm requires a physical boundary condition representing the system's initial state. An anchoring mechanism was engineered by extracting statistics from the first 10 timesteps of each event for every node (e.g., $init_wl_last$, $init_wl_trend$). This acts as a proxy for the initial hidden state.

Crucially for inference reproducibility: If initial water level data is absent or near-zero ($|value| < 10^{-6}$) for specific test nodes, the pipeline employs a strict fallback strategy. The $init_wl_*$ features automatically revert to the historical $node_mean$, while derivative initial features (like $init_wl_trend$ or $init_wl_velocity$) fallback to zero, ensuring inference stability without data leakage.

Variance-Based Sample Weighting

Urban flood datasets are heavily imbalanced, with many nodes remaining dry or static throughout a storm. Tree-based algorithms minimizing Mean Squared Error (MSE) naturally gravitate towards these zero-variance nodes. To counteract this, a variance-based sample weighting strategy was implemented:

$$sample_weight = \frac{1.0}{group_std^2 + 10^{-6}}$$

Crucially, the $group_std$ is computed specifically per $(model_id, node_type)$. This weighting is not an arbitrary choice; it strictly aligns the LightGBM MSE objective with the competition's official Standardized RMSE evaluation metric. By exponentially penalizing errors on highly volatile, hydrodynamically active nodes, this ensures the model directly optimizes exactly what is being evaluated.

3. Physics-Informed Feature Engineering

To overcome the limitations of tree-based algorithms in fluid dynamics, between 70 and 95 features (depending on the specific spatial domain and node type) were engineered to approximate laws of hydraulics, topology, and kinematics.

3.1. Hydraulic & Capacity Proxies

- **Base Topology & Geometry:** Features such as $area$, $calc_depth$, $elevation$, $invert_elevation$, and symmetrical edge features ($area_right$, $roughness_right$) provide 3D spatial context.
- **Surface Topography:** Features including $aspect$, $curvature$, $roughness$, and $surf_slope_max$ define the 2D terrain. Concavity ($curvature$) identifies natural reservoirs.
- **Manning's Formula Approximation:** 1D pipe carrying capacity was approximated using $pipe_dia_mean$, $pipe_slope_mean$, and $pipe_rough_mean$ to derive $total_capacity_proxy$. This enabled dynamic surcharge indicators like $capacity_utilization$ (ratio of short-term rainfall to pipe capacity) and $surcharge_proxy$.

3.2. 1D-2D Spatial Cross-Coupling & Message Passing

To replicate spatial message-passing without GNNs, explicit communication features were engineered between domains:

- **Domain Bridging:** Features such as $coupled_1d_capacity$ and $coupled_2d_flow_acc$ transfer topological constraints across domains.
- **Choke Point Risks:** $cross_choke_risk_1h$ calculates the probability that intense rainfall exceeds a storm drain's intake capacity, while $cross_inlet_load$ models aggregated stress from surface runoff.
- **1-Hop Neighbor Simulation:** Features like $neighbor_amp_max$ and $neighbor_expected_inflow$ transmit the historical hydraulic behavior of adjacent nodes, simulating flow propagation.

3.3. Rainfall Dynamics & Temporal Encoding

- **System Memory:** Exponentially Weighted Moving Averages ($rain_ewm_1h$ to $rain_ewm_72h$) act as the hydrological memory of soil saturation.
- **Rainfall Kinematics & Context:** Derivatives ($rain_acceleration$, $rain_trend_fast$) and global event metrics ($rain_cum$, $rain_cum_pct$, $antecedent_rain$) provide context regarding the storm's phase and intensity.

3.4. Dynamic Node Signatures & Delay Profiles

To prevent overfitting on raw $node_id$s, specific statistical profiles were generated for each location:

- **Historical Statistics:** Baselines like $node_mean$, $node_min$, and $node_max$ define the natural fluctuation bounds. Specifically, the derived feature $node_range = node_max - node_min$ emerged as a top-tier predictor when used in interactions, successfully capturing the absolute historical capacity limits of a node.

- **Sensitivity & Volatility:** $sig_sensitivity$ (water level change per rainfall unit) and $sig_flashiness$ quantify historical responsiveness.
- **Hydraulic Delay Profiling:** $node_avg_delay$ quantifies the lag time between peak rainfall and peak water level. Interactions ($interact_cum_delay$, $interact_rain_delay$) allow the algorithm to temporally shift the predicted hydrograph based on local node properties.

4. Modeling Strategy & Ensemble

The training architecture was designed for physical domain isolation, hyperparameter optimization, and variance reduction.

Domain-Specific Model Isolation

The problem space was partitioned into four independent LightGBM models corresponding to the permutations of urban environments ($model_id$: 1, 2) and topological domains ($node_type$: 1D, 2D). Pressurized fluid dynamics in 1D pipes are governed by different laws than free-surface routing on 2D terrain. Isolating domains prevents feature cannibalization and ensures domain-specific decision tree splits.

Hyperparameter Optimization (Optuna)

Per-domain LightGBM hyperparameters were optimized utilizing the Optuna Bayesian optimization framework. The optimization process used GroupKFold cross-validation (grouped by $event_id$) to prevent temporal leakage. Key optimized parameters included $learning_rate$, num_leaves, max_depth, $feature_fraction$, $bagging_fraction$, and regularization coefficients ($lambda_l1$, $lambda_l2$). The objective function minimized validation RMSE on held-out storm events.

Multi-Seed Bagging & Strategic Architecture Blending

To minimize generalization variance, a multi-seed bagging strategy was implemented. Each domain was trained using three distinct random seeds (42, 888, 2026), with predictions averaged across iterations.

Furthermore, the final submission utilized a 50/50 weighted blend of two distinct tabular architectures to ensure robustness:

- **V182 (Ultimate Signatures):** An aggressive model leveraging complex dynamic node signatures, specifically focusing on the discrepancy between expected hydraulic behavior and actual historical deviations ($sig_expected_vs_actual$).
- **V184 (Purified Engine):** A conservative, strictly filtered iteration focused purely on core hydraulic physics. In this version, overlapping interaction terms and high-risk signature features were explicitly removed to prevent interaction cannibalization and reduce feature-space noise.

Blending these contrasting architectures ensures that localized overconfident predictions in one feature space are dampened by the other.

5. Post-Processing: Enforcing Physical Constraints

While gradient boosting algorithms capture non-linear interactions efficiently, they construct predictions using orthogonal splits, resulting in discontinuous, step-wise output functions. Two post-processing steps were implemented to enforce physical reality.

Physical Wave Continuity (Savitzky-Golay Filter)

Urban water routing is a continuous kinematic process. Uncorrected LightGBM predictions exhibited high-frequency step artifacts. To restore physical continuity, a Savitzky-Golay filter ($window_length = 5, polyorder = 2$) was applied to the final blended predictions.

Unlike a Simple Moving Average (SMA), which severely attenuates sharp signals (detrimental to RMSE in flash flood scenarios), the Savitzky-Golay filter operates by fitting a low-degree polynomial via linear least squares. This effectively smoothed out decision-tree step artifacts while strictly preserving the timing and maximum amplitude of the flood peaks.

Physical Bounding (Minimum Depth Clipping)

To prevent the model from predicting physically impossible negative water depths due to residual subtraction, a strict lower bound was enforced. All final reconstructed predictions were clipped from below using the historical minimum water level of the respective node:

$$pl.max_horizontal(pl.col("node_mean") + pl.col("pred_residual"), pl.col("node_min"))$$

6. Conclusion and Reproducibility

This solution demonstrates that Physics-Informed Tabular Machine Learning can compete with complex deep learning architectures in spatial-temporal flood routing tasks. By embedding explicit hydraulic constraints, classical gradient boosting algorithms were forced to generalize fundamental laws of mass conservation rather than memorize static infrastructure layouts. The resulting architecture demonstrated high interpretability, immunity to autoregressive error accumulation, and exceptional robustness to structural dataset anomalies.

Computational Profile

While avoiding deep learning architectures bypasses the need for GPU clusters, this complex tabular approach requires substantial RAM bandwidth and CPU multi-threading. Generating between 70 and 95 features across millions of timesteps and executing multi-level spatial cross-joins creates a heavy system load. Training the complete pipeline required approximately 4 hours of intensive computation on a local workstation (16 physical cores, 32 threads, 64 GB RAM).

However, this is a one-time training cost. During the inference phase, generating predictions using the ensemble of saved LightGBM trees is computationally efficient and low-latency, parallelizing effectively on standard consumer CPUs, suggesting high potential applicability for operational flood forecasting systems.